

Experiment No.	2
Experiment Title	Diabetes Prediction using Logistic Regression ¹
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1 Objective

To build a machine learning pipeline that predicts whether a patient has diabetes from basic clinical attributes, covering EDA, preprocessing, feature scoring, a stratified train-validation-test split, and classification using Logistic Regression.

2 Problem Statement

Given a patient's demographic and clinical record (gender, age, hypertension, heart disease, smoking history, BMI, HbA1c level, blood glucose level), predict the binary target **diabetes** (1 = diabetic, 0 = non-diabetic). This is useful as a low-cost first-pass screening tool before confirmatory lab tests.

3 Dataset Description

Dataset Name	Diabetes Prediction Dataset
Dataset Source	<code>diabetes_prediction_dataset.csv</code>
Number of Samples	100,000
Number of Features	8 (excluding target diabetes)
Number of Classes	2 (0 = No Diabetes, 1 = Diabetes)
Missing Values	None
Train-Validation-Test Split	70% train (70,000) – 15% validation (15,000) – 15% test (15,000), stratified, <code>random_state=42</code>

4 Exploratory Data Analysis

The dataset has 7 numeric columns (`age`, `hypertension`, `heart_disease`, `bmi`, `HbA1c_level`, `blood_glucose_level`, `diabetes`) and 2 categorical columns (`gender`, `smoking_history`). There are no missing values, but 3,854 duplicate rows were found in the raw data.

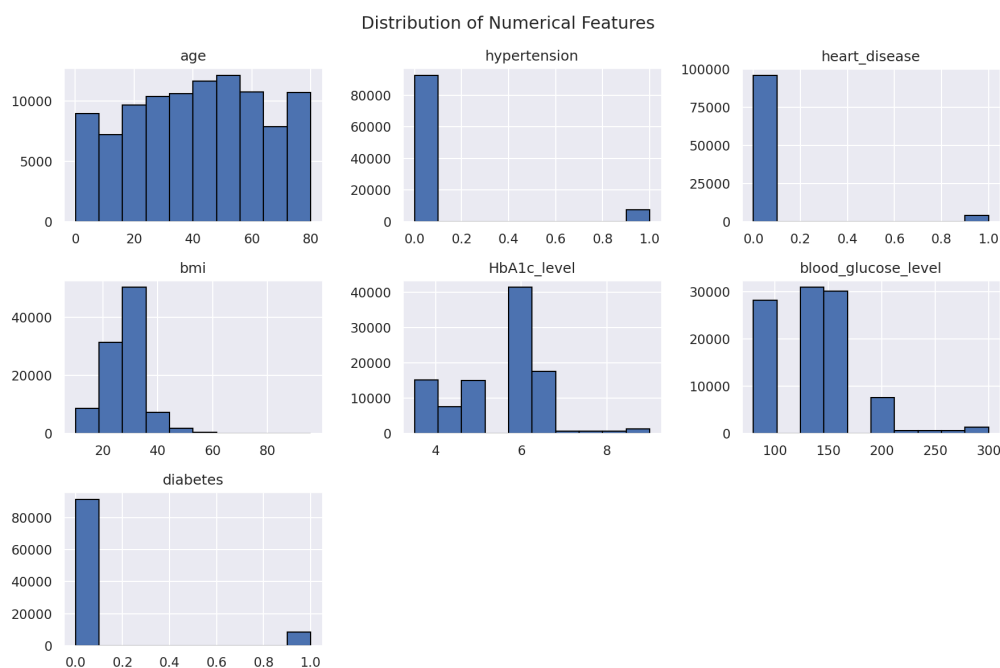


Figure 1: Distribution of numerical features.

Inference: Most patients are non-hypertensive and free of heart disease (the corresponding histograms are heavily skewed toward 0). `age` and `bmi` look close to normally distributed, while `HbA1c_level` and `blood_glucose_level` show a few distinct clusters rather than a smooth curve, which suggests the values may have been recorded/binned from discrete lab test readings. The `diabetes` histogram itself confirms a strong class imbalance.

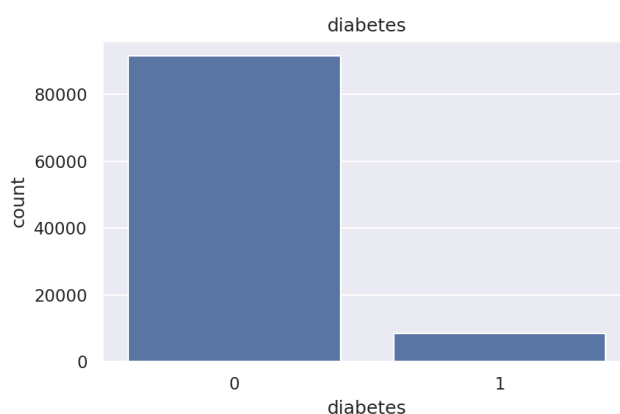


Figure 2: Class distribution of the target variable `diabetes`.

Inference: Only 8,500 of the 100,000 patients (8.5%) are labeled diabetic, against 91,500 non-diabetic. This is a significant imbalance, so accuracy alone will be a misleading metric – precision, recall and F1-score on the minority (diabetic) class matter far more here.

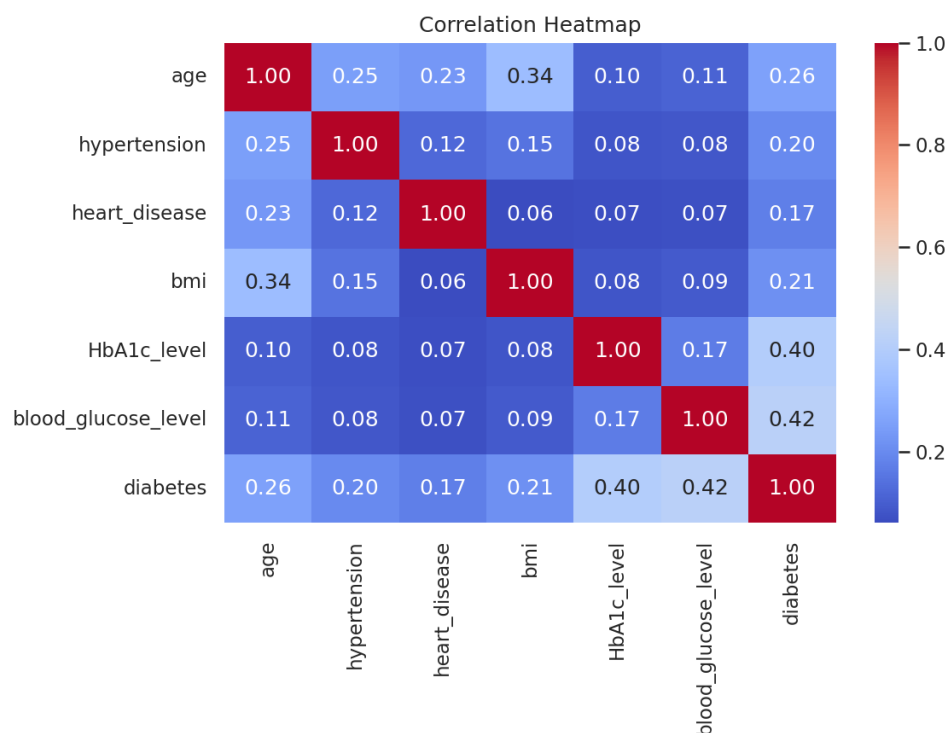


Figure 3: Correlation heatmap of numeric features.

Inference: Correlations among the raw numeric features are generally weak, with `blood_glucose_level` and `HbA1c_level` showing the strongest (though still moderate) association with `diabetes`. No pair of predictor features is strongly collinear, so multicollinearity is not a major concern for the logistic regression model.

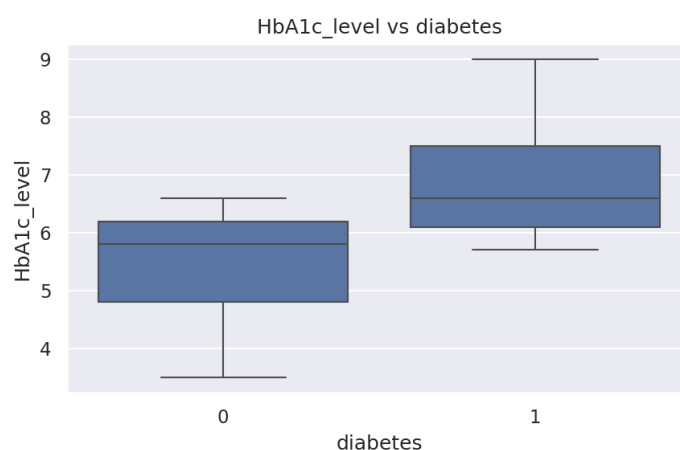


Figure 4: HbA1c level across diabetes classes.

Inference: Diabetic patients (class 1) show a visibly higher median HbA1c level than non-diabetic patients, with limited overlap between the two boxes. This is clinically expected – HbA1c is a direct marker of long-term blood sugar control – and explains why it receives a high ANOVA score during feature selection.

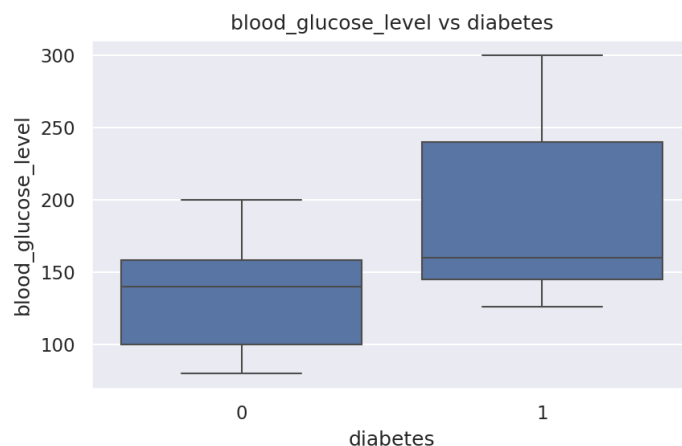


Figure 5: Blood glucose level across diabetes classes.

Inference: Similarly, blood glucose level is clearly higher for diabetic patients, with a wider spread and higher median. Together, Figures 4 and 5 explain why these two features top both the ANOVA and Chi-Square rankings in Section 6.

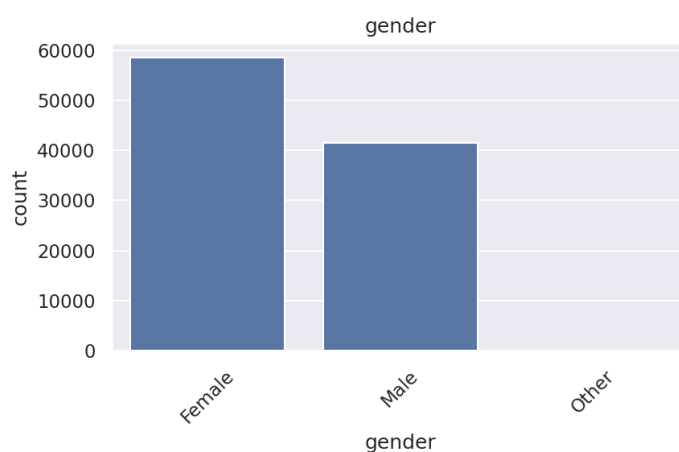


Figure 6: Distribution of the `gender` attribute.

Inference: The dataset is skewed toward female patients, with a small number of “Other” entries. Gender later receives the lowest ANOVA/Chi-Square score among all features, suggesting it is a weak individual predictor of diabetes in this data.

5 Data Preprocessing

- **Missing values:** Handled via `SimpleImputer` (median for numeric columns, most-frequent for categorical columns); this dataset had none, so imputation had no visible effect, but the step keeps the pipeline robust to future data.
- **Encoding:** `gender` and `smoking_history` were label-encoded to integers.

- **Feature scaling:** All 8 features were standardized with `StandardScaler` (zero mean, unit variance), which logistic regression’s gradient-based solver converges faster with.
- **Train-validation-test split:** `train_test_split` was applied twice (stratified on `diabetes`) to produce a 70/15/15 split, preserving the 91.5:8.5 class ratio in every subset.
- **Duplicates:** Not removed in this pipeline – the 3,854 duplicate rows remain in the data before splitting, a point raised in the Discussion.

6 Feature Selection

ANOVA F-test and Chi-Square scores were computed for all 8 features (Chi-Square requires non-negative inputs, so any negative-valued column would be shifted before scoring):

Table 2: Feature scores, ranked by ANOVA F-score.

Feature	ANOVA Score	Chi-Square Score
<code>blood_glucose_level</code>	21363.05	211290.67
<code>HbA1c_level</code>	19122.21	3329.14
<code>age</code>	7131.41	80576.72
<code>bmi</code>	4816.11	7407.90
<code>hypertension</code>	4072.71	3620.49
<code>heart_disease</code>	3038.56	2832.76
<code>smoking_history</code>	897.02	1456.49
<code>gender</code>	140.15	82.05

`blood_glucose_level` and `HbA1c_level` dominate both rankings, consistent with the boxplots in Section 4. Unlike a hard top- k selection, this pipeline uses the scores only for interpretation – all 8 standardized features were retained for training the model.

7 Machine Learning Algorithm: Logistic Regression

Working principle: Logistic regression models the probability of the positive class using the sigmoid function applied to a linear combination of the input features.

Mathematical formulation:

$$P(y = 1 \mid x) = \sigma(w^\top x + b) = \frac{1}{1 + e^{-(w^\top x + b)}}$$

The weights w and bias b are learned by minimizing the log-loss (binary cross-entropy):

$$J(w, b) = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right]$$

using an iterative solver (`lbfgs` in this experiment).

Advantages: Fast to train, gives probabilistic outputs, coefficients are directly interpretable as log-odds effects, and works well when classes are close to linearly separable in feature space.

Limitations: Assumes a linear decision boundary in the (scaled) feature space, is sensitive to class imbalance unless reweighted, and does not automatically capture feature interactions.

8 Experimental Setup

Python Version	3.12.3
Scikit-Learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Linux (Ubuntu)
Hardware	Standard CPU-only environment

9 Hyperparameter Selection

Parameter	Value
max_iter	1000
solver	lbfgs (default)
penalty	l2 (default)
C (inverse regularization strength)	1.0 (default)
random_state	42

No hyperparameter search was performed; the default `LogisticRegression(max_iter=1000, random_state=42)` configuration was used to establish a baseline.

10 Results

Metric	Value
Accuracy	0.9597
Precision	0.8594
Recall	0.6282
F1-score	0.7259
ROC-AUC	0.9599

On the 15,000-sample test set, the model correctly classified 14,395 patients (13,594 true negatives, 801 true positives), with 131 false positives and 474 false negatives.

11 Result Visualization

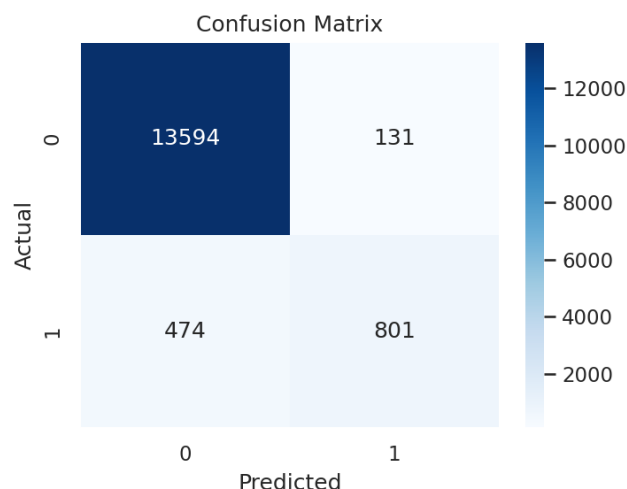


Figure 7: Confusion matrix for the Logistic Regression classifier.

Inference: The model is very accurate on the majority (non-diabetic) class but misses 474 of 1,275 actual diabetic patients (recall $\approx 63\%$). In a screening context, these false negatives are the more costly error, so the default 0.5 probability threshold may not be the right operating point.

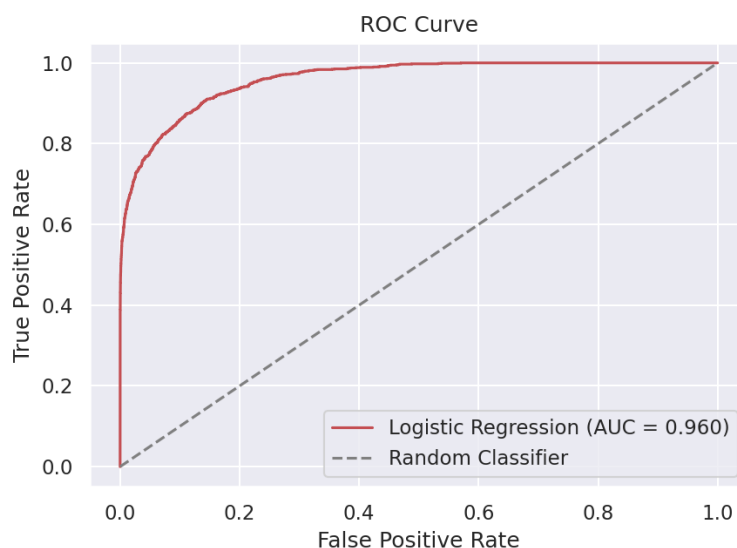


Figure 8: ROC curve for the Logistic Regression classifier.

Inference: A ROC-AUC of 0.96 shows the model separates the two classes well when ranked by predicted probability, even though the default threshold underperforms on recall – confirming that threshold tuning (rather than the model itself) is the main lever for improving diabetic-case detection.

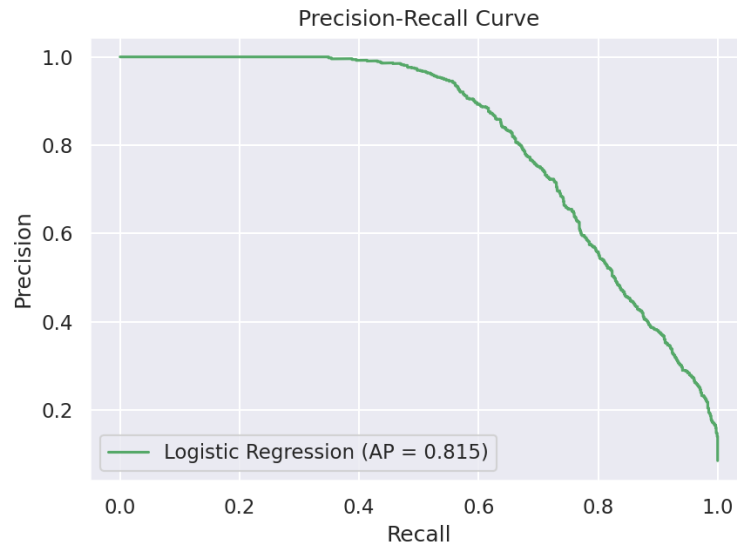


Figure 9: Precision-Recall curve for the Logistic Regression classifier.

Inference: The PR curve drops off more visibly than the ROC curve as recall increases, which is expected given the 91.5:8.5 class imbalance – it is a more sensitive diagnostic here and shows that pushing recall higher will come at a real precision cost unless the model or sampling strategy is improved.

12 Discussion

The Logistic Regression model achieved 95.97% accuracy with a ROC-AUC of 0.96, indicating good overall performance for diabetes prediction. Blood glucose level and HbA1c level were identified as the most influential features. Although the model performed well, the recall for the diabetic class was lower due to class imbalance, resulting in some diabetic cases being missed. Future improvements can include class balancing techniques, hyperparameter tuning, and testing other machine learning models to further improve recall.

13 Conclusion

This experiment successfully implemented a complete machine learning pipeline for diabetes prediction, including data preprocessing, feature selection, model training, and evaluation using Logistic Regression. The model achieved an accuracy of 95.97% and a ROC-AUC of 0.96, demonstrating strong classification performance. Feature selection showed that blood glucose level and HbA1c level are the most important predictors of diabetes. Overall, the model provides reliable prediction performance and can serve as a useful baseline for diabetes risk assessment, while future improvements can focus on handling class imbalance and optimizing model parameters.

14 References

- [1] Chitraju Vishnu Vineeth, “Diabetes Prediction ML Workflow,” GitHub Repository. [Online]. Available: https://github.com/VishnuVineeth14/ML_LAB_ASSIGNMENTS/blob/main/Assignment1/Diabetes_Prediction_ML_Workflow.ipynb. [Accessed: Jul. 2026].